Build: from a clean machine to a working product

English version of ../BUILD.md.

Pinned versions

Tool	Version	Source of truth
Python	3.11	backend/requirements.txt, ci.yml, scripts/doctor.sh
Node.js	18+ (frontend build)	frontend/package.json, ci.yml
PostgreSQL	16	docker-compose.yml, infra/yandex-cloud/
Python dependencies	exact versions (==)	backend/requirements.txt
npm dependencies	lockfile	frontend/package-lock.json (npm ci)
Alembic	1.14	the revision in backend/alembic/versions

Level A — source code

Locally only Python 3.11 is needed: the tests run on SQLite.

```
make doctor     # tools, .env, ports; changes nothing
make install
make test       # pytest, 23 files
make lint       # ruff
make typecheck  # mypy
make alembic-check
make docs-check
```

Level B — artefacts

```
make build-frontend   # frontend/dist
```

The backend artefact is the image built from backend/Dockerfile.

Level C — integration environment

```
cp .env.example .env    # SECRET_KEY, MINIO_ROOT_PASSWORD,
                        # KEYCLOAK_ADMIN_PASSWORD, DATABASE_URL are mandatory
make compose-up
make alembic-upgrade DATABASE_URL='postgresql+asyncpg://...'
make seed               # optional
make smoke-test
```

A minimal set of services (without Keycloak/Redis/MinIO):

```
make compose-up SERVICES="backend frontend nginx"
```

Level D — container build

```
docker compose build
docker compose up -d
./scripts/smoke-test.sh http://localhost:8000 http://localhost:3000
```

Level E — clean deployment

```
docker compose down -v          # deliberately: deletes data
make compose-up
make alembic-upgrade DATABASE_URL='...'   # schema from scratch
pg_dump → pg_restore → make smoke-test    # verifying a copy
```

Automated in `.github/workflows/portability.yml`.

Environment variables

The full list is in `.env.example`. Mandatory (without them compose does not start): `SECRET_KEY`, `MINIO_ROOT_PASSWORD`, `KEYCLOAK_ADMIN_PASSWORD`, `DATABASE_URL`. Optional integrations: `GIGACHAT_CREDENTIALS`, `KEYCLOAK_*`, `S3_*`, `NOTIFICATIONS_*`.

Typical errors

Error	Cause	Solution
`DATABASE_URL must be set`	empty `.env`	fill in the mandatory values
`/readyz` is not 200	the database is unreachable or migrations are not applied	`make alembic-current`, backend logs
`Target database is not up to date`	the revision is behind	`make alembic-upgrade`
Tests "pollute" the dev database	an exported `DATABASE_URL`	`conftest.py` forces `test.db` — do not override it
The frontend cannot see the API	`VITE_API_URL`/CORS	see `frontend/.env.example`, `CORS_ORIGINS`
`npm ci` fails	the lockfile is out of sync	change `package.json` and the lockfile together

Proof

`ci.yml` — levels A–B, `portability.yml` — levels C–E. The matrix is in docs/verification/matrix.md.